

Question

It is known that for H_2S , $K_{\text{a}1} = 1.1 \times 10^{-7}$, $K_{\text{a}2} = 1.3 \times 10^{-13}$; $K_{\text{sp}}(\text{CuS}) = 6.3 \times 10^{-36}$, $K_{\text{sp}}(\text{Cu}_2\text{S}) = 2.5 \times 10^{-48}$. $\varphi_{\text{A}}^{\theta}(\text{S}/\text{H}_2\text{S}) = 0.142 \text{ V}$, $\varphi_{\text{B}}^{\theta}(\text{S}/\text{S}^{2-}) = -0.476 \text{ V}$; $\varphi_{\text{A}}^{\theta}(\text{NO}_3^-/\text{NO}) = 0.942 \text{ V}$, $\varphi^{\theta}(\text{Cu}^{2+}/\text{Cu}) = 0.342 \text{ V}$, $\varphi^{\theta}(\text{Cu}^+/\text{Cu}) = 0.521 \text{ V}$.

Based on the above conditions, write the ionic equation for the dissolution of Cu_2S in dilute nitric acid to form a blue solution and a yellow precipitate, and calculate the standard equilibrium constant of this reaction using Hess's law. Select the option closest to your calculated equilibrium constant.

A. All other options are incorrect

B. $K^{\theta} = 1.1 \times 10^{80}$

C. $K^{\theta} = 2.0 \times 10^{130}$

D. $K^{\theta} = 2.4 \times 10^8$

E. $K^{\theta} = 5.1 \times 10^{43}$

Answer

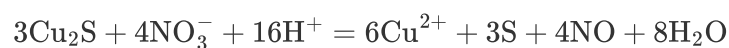
Correct Answer: **B**

Detailed Explanation

First, calculate the standard potential for the reduction of cupric ion to cuprous ion:

$$\varphi_B^\theta(\text{Cu}^{2+}/\text{Cu}^+) = 2\varphi^\theta(\text{Cu}^{2+}/\text{Cu}) - \varphi^\theta(\text{Cu}^+/\text{Cu}) = 0.163 \text{ V}$$

The chemical equation mentioned in the question is:



CHECKPOINT

1 PTS

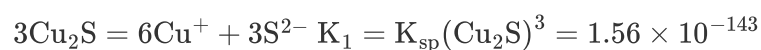
The chemical equation is: $3\text{Cu}_2\text{S} + 4\text{NO}_3^- + 16\text{H}^+ = 6\text{Cu}^{2+} + 3\text{S} + 4\text{NO} + 8\text{H}_2\text{O}$

Decompose the reaction to separate the precipitation dissolution equilibrium and the redox reaction:

CHECKPOINT

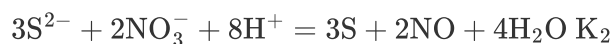
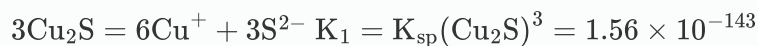
1 PTS

Decompose the reaction to separate the precipitation dissolution equilibrium and the redox reaction



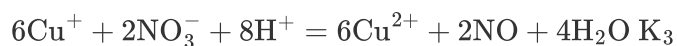
CHECKPOINT

1 PTS



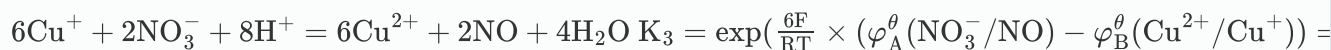
CHECKPOINT

2 PTS



CHECKPOINT

2 PTS



Where, for K_2 and K_3 , we have

$$K_2 = \exp\left(\frac{6F}{RT} \times (\varphi_A^\theta(\text{NO}_3^-/\text{NO}) - \varphi_B^\theta(\text{S}/\text{S}^{2-}))\right) = 6.64 \times 10^{143}$$

$$K_3 = \exp\left(\frac{6F}{RT} \times (\varphi_A^\theta(\text{NO}_3^-/\text{NO}) - \varphi_B^\theta(\text{Cu}^{2+}/\text{Cu}^+))\right) = 1.03 \times 10^{79}$$

$$K^\theta = K_1 K_2 K_3 = 1.1 \times 10^{80}$$

CHECKPOINT

1 PTS

$$K^\theta = K_1 K_2 K_3 = 1.1 \times 10^{80}$$